Idenso

Symbolic tensor identities for Spenso and Symbolica

---

1 Overview

Idenso is the symbolic identity and simplification layer for tensors encoded in the form that Spenso can parse from Symbolica expressions. It provides representation symbols, index tooling, reversible "cooking" of subexpressions, selective expansion, and identities for Dirac, metric, epsilon, and color algebra.

> **Symbolic, not component expansion**
> Idenso rewrites abstract tensor expressions. Functions such as `expand_mink` distribute factorized terms that carry selected index families; they do not turn every tensor into a dense array of components. Concrete tensor storage and network execution belong to Spenso.

1.1 Choose a task

- To install Symbolica and simplify one metric contraction with its bundled community modules, follow the Python guide.
- To apply the same identity through the native API, use the Rust guide.
- To verify the rewrite as a controlled, observable pass, follow the controlled identity tutorial and the Python function reference.
- To isolate dummy-index namespaces or cook a large expression, use the algebra guide with the exact `IndexTooling` and `Cookable` Rustdoc.
- To audit a sign or normalization in a Dirac/color pass, use the source-backed shipped color and Dirac convention reference.

1.2 A controlled rewrite pipeline

There is no universally correct "simplify everything" order. A robust workflow makes each phase explicit:

- initialize the standard representations and tensor symbols;
- inspect dangling indices and normalize or wrap dummy-index namespaces when combining expressions;
- expand only the sectors needed by the next identity pass;
- simplify gamma, metric, or color structures as appropriate;
- canonicalize and compare results using the conventions of the consuming calculation.

Expansion can grow an expression quickly, so perform it as late and selectively as possible. Wrapping indices is especially important before multiplying independently constructed expressions: equal printed index names can otherwise acquire an unintended contraction.

1.3 Representations and cooking

The representation layer defines spin-fundamental, color-fundamental, color-sextet, bispinor, and color-adjoint types together with their duality. Importing the published Idenso community module registers the related Symbolica symbols and Spenso tags before expressions are built.

The `Cookable` API can replace selected function-like subexpressions or index payloads with compact symbols and later reverse the operation. Use reversible settings when the original expression must be recovered. A cooked symbol is an intermediate encoding, not a portable serialized physics result unless the associated mapping and settings are retained.

Idenso is consumed by GammaLoop and its tensor conventions are shared with Spenso. Continue to the interface guide, native Rustdoc, and generated Python reference for supported functions, signatures, and feature gates.

Contributors changing representation syntax, symbolic-network parsing, index ownership, or rewrite order should also read the source-audited Idenso implementation architecture.

2 Simplify your first tensor expression

Idenso supplies representation-aware identities through both Symbolica's Python community module and a native Rust crate. Choose the interface that already owns your symbolic expressions; both routes simplify a Minkowski metric contraction while preserving its one free index.

Python
Use the bundled community module for notebooks, exploratory algebra, and Python pipelines. This is the shortest supported installation path.

Use Idenso from Python →

Rust
Use the crate when a Rust application owns Symbolica expressions and should compose Idenso's metric, Dirac, color, or cooking passes directly.

Use Idenso from Rust →

The interface guide maps both routes to the same identity families and explains their feature and registration boundaries.

3 Using Idenso from Python

Symbolica `2.2.0` bundles Idenso and Spenso as native community modules. This example contracts one four-dimensional Minkowski metric with a vector and verifies the exact remaining expression.

3.1 Install and verify the modules

```
python -m venv .venv
. .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install "symbolica==2.2.0"
python -c "import symbolica.community.idenso; import symbolica.community.spenso"
```

There is no standalone `idenso` Python wheel. Import the community module before constructing expressions, and follow Symbolica's installation and license terms.

3.2 Simplify one metric contraction

Save this as `idenso_quickstart.py`:

```
from symbolica.community.idenso import initialize, list_dangling, simplify_metrics
from symbolica.community.spenso import Representation, TensorName

initialize()
rep = Representation.mink(4)
mu, nu = rep("mu"), rep("nu")
g, q = TensorName.g(), TensorName("q")
```

```
expression = g(mu, nu) * q(mu)
reduced = simplify_metrics(expression)

assert reduced == q(nu).to_expression()
assert len(list_dangling(reduced)) == 1
print(reduced)
```

Run `python idenso_quickstart.py`. Success means the explicit metric disappears and `q(nu)` remains. The structural equality check is stronger than comparing printed text, whose formatting may change between Symbolica releases.

> **Make rewrite stages observable**
> Check dangling indices before and after each pass. When multiplying expressions built with independent dummy-index namespaces, wrap those namespaces before simplification so equal printed names do not create an accidental contraction.

Continue with the controlled identity tutorial for a verification workflow, then use the algebra guide to compose metric, Dirac, color, and cooking stages deliberately.

4 Using Idenso from Rust

This route constructs the same indexed metric contraction as the Python example, applies Idenso's native `MetricSimplifier`, and checks the exact reduced Symbolica expression.

> **Use the current Git API until the next crate release**
> The published `idenso 0.3.0` predates the current module layout and uses an older Symbolica dependency. Mixing its API with the current manual can create incompatible `Atom` types. The pinned Git dependency below matches this documentation; switch to the next published Idenso version once it carries the same API.

4.1 Create a Rust project

Use Rust 1.96 or newer:

```
cargo new idenso-quickstart
cd idenso-quickstart
cargo add idenso \
  --git https://github.com/alphal00p/gammaloop.git \
  --rev 875114f143442b543d4795b5c29586bc6f40f2e6
cargo add symbolica@3.0.1 --no-default-features
```

Replace `src/main.rs` with:

```
use idenso::shorthands::metric::MetricSimplifier;
use symbolica::parse_lit;

fn main() {
    idenso::representations::initialize();

    let expression = parse_lit!(
        spenso::g(spenso::mink(4, 0), spenso::mink(4, 1))
            * p(spenso::mink(4, 1))
    );
    let reduced = expression.simplify_metrics();

    assert_eq!(reduced, parse_lit!(p(spenso::mink(4, 0))));
```

```
    println!("{reduced}");
}
```

Run `cargo run`. Success means the assertion passes and the vector retains index `0` after the metric contracts index `1`. Symbolica's license terms apply to the native crate as well as the Python module.

Continue with the algebra guide before composing multiple identity families or multiplying expressions with independently allocated dummy indices.

5 Tutorial

This tutorial uses Idenso's Python community module to contract one four-dimensional Minkowski metric tensor with a vector. It is intentionally small: the point is to establish the registered tensor syntax and observe one algebra pass before composing a larger Dirac or color pipeline.

For the shortest installation and first-run path, begin with Using Idenso from Python. Then return here to inspect the identity pass and extend it into an algebra pipeline.

5.1 Prerequisites

Idenso is mounted at `symbolica.community.idenso`; it is not installed by `pip install idenso`. Install the published Symbolica distribution, which bundles the Idenso and Spenso community modules, and verify the actual environment first:

```
python -m pip install --upgrade symbolica
python -c "import symbolica.community.idenso; import symbolica.community.spenso"
```

If that fails, check the installed Symbolica version before continuing. Generating a `.pyi` file does not mount the native module. Source embedders can build a custom community assembly.

5.2 Simplify one metric contraction

Save the following as `metric_first.py`:

```
from symbolica.community.idenso import list_dangling, simplify_metrics
from symbolica.community.spenso import Representation, TensorExpression, TensorName

rep = Representation.mink(4)
mu = rep("mu")
nu = rep("nu")

g = TensorExpression.g(rep)
q = TensorName("q")
expression = g(mu, nu) * q(mu)

free_before = list_dangling(expression)
reduced = simplify_metrics(expression)
free_after = list_dangling(reduced)

assert len(free_before) == 1
assert len(free_after) == 1
print("reduced:", reduced)
```

Run `python metric_first.py`. Success means both rank assertions pass, the metric is removed from the reduced expression, and the result is a rank-one expression carrying `nu`. The rewrite is abstract: the registered Minkowski metric supplies index compatibility, but this example does not substitute a numerical signature. Printed output can vary with the installed Symbolica version; inspect the expression structure instead of comparing exact text.

Verification scope and cost
The docs harness compiles this Python source without importing native modules; it syntax-checks the environment probe without running it. Execute both steps in a provisioned community-module environment to verify the rank-one invariant. The script should take seconds after startup; building or installing Symbolica community modules is the expensive prerequisite and may take substantially longer.

Import before constructing expressions
Importing `symbolica.community.idenso` registers Idenso's representation and tensor symbols. Construct the Spenso-compatible expression after that import so Idenso's matchers see the intended slots and tensor names.

5.3 Grow this into an algebra pipeline

Keep transformations observable while developing:

- call `list_dangling` before and after a pass to catch accidental contractions;
- use `wrap_dummies` before multiplying expressions built in independent index namespaces;
- expand only the Minkowski, bispinor, or color sector needed by the next pass;
- apply `simplify_metrics`, `simplify_gamma`, and `simplify_color` as distinct phases;
- canonicalize only after the physics-specific identities required by the calculation are explicit.

5.4 Troubleshooting and next steps

- `ModuleNotFoundError` for `symbolica.community.idenso` means the installed Symbolica build does not include the native module; a `.pyi` type stub or source checkout alone is not enough.
- If the metric remains unchanged, construct its slots with the same `Representation` and abstract index objects as the vector. Plain Symbolica functions do not automatically carry Spenso tensor structure.
- If a free index disappears unexpectedly, inspect repeated names and use `wrap_dummies` before combining separately constructed expressions.
- Continue with the syntax-and-algebra manual, then use the Python and Rust API references for signatures and feature gates.

6 Syntax, indices, and algebra passes

Idenso consumes the Symbolica function form emitted for Spenso structures. Function heads carry representation meaning and their arguments carry abstract indices or tensor data. Import the Idenso community module before parsing so Symbolica attributes, Spenso tags, and dual representations are registered in one deterministic order.

6.1 Indices and cooking

Free indices describe the result; repeated compatible indices describe contractions. Before combining independently built expressions, wrap or rename dummy-index namespaces so equal printed names do not create accidental contractions. Cooking temporarily replaces selected index payloads or function subexpressions with compact symbols. Retain the generated map and settings whenever uncooking is required.

Cooking is especially useful before expensive canonicalization, but it changes what downstream matchers can see. Uncook before applying an identity whose pattern depends on the hidden tensor head or index structure.

6.2 Keep independent dummy namespaces independent

Two factors may legitimately print the same local dummy name before they are multiplied. Wrap each factor with a distinct header first, while leaving its external indices untouched:

```python
import symbolica as sp
from symbolica.community.idenso import list_dangling, wrap_dummies
from symbolica.community.spenso import Representation, TensorExpression, TensorName

rep = Representation.euc(3)
mu = rep("mu")
nu = rep("nu")
rho = rep("rho")
g = TensorExpression.g(rep)
p = TensorName("p")
q = TensorName("q")

left = g(mu, nu) * p(mu)
right = g(mu, rho) * q(mu)
safe_product = wrap_dummies(left, sp.S("lhs")) * wrap_dummies(right, sp.S("rhs"))

assert len(list_dangling(safe_product)) == 2
```

The stable invariant is two free indices, `nu` and `rho`; the two occurrences of local `mu` belong to separate contractions after wrapping. The generated `wrap_dummies` reference records the Python signature, while the exact `IndexTooling` Rustdoc covers the underlying Rust boundary.

> **Interpret index failures before simplifying**
> More or fewer than two dangling indices means a name collided or a slot's representation or duality differs from the intended one. An unchanged plain Symbolica function means it was not constructed through Spenso tensor names/representations, or the Idenso module was imported only after parsing. Correct those structural issues before metric, Dirac, or color simplification.

6.3 Metric and epsilon operations

Metric contraction raises, lowers, or identifies compatible Lorentz indices according to the registered representation. Epsilon identities depend on dimension, ordering, and sign conventions; expand them only when the next pass benefits from the larger expression. Keep Minkowski expansion selective to avoid distributing unrelated scalar factors.

6.4 Dirac and color algebra

Dirac passes simplify gamma chains, traces, slashes, and spinor-compatible contractions. Color passes handle fundamental/adjoint deltas, generators, structure constants, and registered group parameters. Apply one algebra family at a time and inspect the intermediate expression; an all-at-once fixed-point loop can obscure which convention produced a sign or normalization.

> **Canonical does not mean physically equivalent by itself**
> Canonical ordering makes structurally equivalent expressions comparable under the registered rules. On-shell relations, gauge choices, dimension-specific identities, and model parameter substitutions must still be requested explicitly.

6.5 Schoonschip network parsing

The Schoonschip path parses large gamma-chain expressions into the same Spenso-compatible network model before contraction. It resolves aliases and normalizes the expression before constructing the network. Spenso then plans and executes contractions, while Idenso supplies the representation and algebra rules. Avoid substituting scalar parameters too early: doing so can substantially increase intermediate expression size even when the resulting tensor network is unchanged.

Concrete syntax and rewrite cases are documented in the Spenso/Symbolica syntax note and the rendered shipped color and Dirac convention reference. The Schoonschip parsing guide shows how normalization, network construction, and contraction fit together.

Tensor storage and execution remain owned by Spenso.

7 Shipped color and Dirac rules

This page states the conventions implemented by Idenso's public simplifiers. It is deliberately shorter than the implementation source map: a physicist should be able to decide whether a rule applies without reading FORM internals, proposed patterns, or historical validation commands.

7.1 Input and scope

Idenso rewrites representation-aware Spenso expressions after `initialize()`. A printed function that merely resembles `gamma`, `t`, or `f` is not enough: its Lorentz, bispinor, fundamental, and adjoint slots must carry the registered representations and compatible dimensions. Unrecognized or unsupported structures remain in the expression rather than acquiring tensor meaning from their names.

The public Python entry points use the default `GammaSimplifySettings` and `ColorSimplifySettings`. Rust callers can choose other settings, so record those settings when a normal form is part of a reproducible result.

7.2 Dirac convention

For compatible Lorentz and bispinor representations, Idenso uses the Clifford relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}1.$$

The metric object and its signature belong to the registered Spenso representation; the simplifier does not silently replace it by a numerical diagonal metric. Ordinary gamma chains support adjacent contractions and odd/even trace recursion in a dimension-compatible form. In the explicit four-dimensional representation, the checked trace identities include

$$\operatorname{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu},$$

$$\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho) = 0,$$

and

$$\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}).$$

Chisholm reductions, gamma-zero conjugation, chiral projectors, and gamma-five identities are strictly four-dimensional in the implementation. In particular, the shipped gamma-five trace convention is

$$\operatorname{tr}(\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4\varepsilon^{\mu\nu\rho\sigma}.$$

This equation fixes Idenso's internal epsilon normalization: it contains no additional factor of i. It is not a prescription for gamma five in dimensional regularization. Expressions using a specific scheme such as 't Hooft–Veltman or Larin require an explicit scheme-aware conversion outside this default simplifier. The optional three-gamma epsilon expansion exists in the Rust settings but is disabled by the default Python entry point.

The maintained runtime cases are the FORM-reference Dirac tests and the dimension-gating tests.

7.3 Color convention

Fundamental generators are written $(T^a)_i^{\ j}$. Idenso keeps the trace normalization T_R and Casimirs symbolic unless an explicit Rust conversion setting specializes them. The implemented normalization is

$$\operatorname{tr}\big(T^a T^b\big) = T_R\delta^{ab},$$

$$\sum_a (T^a)^j_i (T^a)^l_k = T_R\left(\delta^l_i \delta^j_k - N_c^{-1}\delta^j_i \delta^l_k\right),$$

$$\sum_a (T^a)^j_i (T^a)^k_j = C_F \delta^k_i,$$

and

$$\sum_{c,d} f^{acd} f^{bcd} = C_A \delta^{ab}.$$

For the conventional fundamental representation of $\mathrm{SU}(N_c)$, one may subsequently choose $T_R = \frac{1}{2}$, $C_A = N_c$, and $C_F = \frac{N_c^2-1}{2N_c}$. Those substitutions are a specialization, not assumptions imposed on every expression by `simplify_color`.

The default color pass evaluates supported closed traces and expands supported cross-chain fundamental Fierz contractions. Open lines, higher invariant tensors, or unsupported structures may remain explicit in the result; that is a partial normal form, not an exception. The normalization and free/dummy-index placement above are locked by the FORM-reference color tests.

7.4 What is shipped

Rule family	Status	Boundary
Ordinary Clifford chains and traces	Shipped	Compatible dimensions; default Python path.
Chisholm, gamma zero, projectors, and gamma five	Shipped	Explicit four-dimensional representations only.
Three-gamma epsilon expansion	Opt-in	Rust `GammaSimplifySettings`; disabled by the Python default.
Color traces, Casimirs, structure contractions, and fundamental Fierz	Shipped	Registered color representations; invariants remain symbolic by default.
Every FORM `color.h` identity and high-order invariant	Not implied	Only rules represented in the current Idenso implementation and tests are public behavior.

> **Do not mix algebraic and physical assumptions**
> Idenso's canonical form records what follows from its registered algebra rules. On-shell relations, gauge choices, external-state sums, dimensional-regularization schemes, and model substitutions are separate assumptions and must be applied explicitly.

The implementation-oriented source mapping, FORM excerpts, target patterns, historical commands, and unresolved provenance questions are retained in the Idenso FORM color/Dirac source map. They are contributor evidence, not additional claims about the public runtime surface.

8 Rust and Python APIs

8.1 Rust package

The `idenso` crate exposes representation types and syntax macros together with several rewrite families:

- `IndexTooling` covers canonicalization, conjugation, index wrapping, and dangling-index inspection for Symbolica atoms;
- `Cookable`, `CookSettings`, and the cook filters control reversible or flattening encodings;
- `SelectiveExpand` expands terms by recognized representation families;
- `dirac`, `color`, `epsilon`, and shorthand modules implement algebra-specific rewrites;
- `representations::initialize` installs the standard representation and tensor symbols.

Representation helper macros such as `bis!`, `cof!`, and `coad!` construct the symbolic forms expected by Spenso and Idenso. The Rust orientation leads to the revision-specific Rustdoc for their accepted forms, return types, feature gates, and source locations. Representation encoding is always available. APIs behind `reference-cases`, `python`, and `python-stubs` require the matching Cargo feature.

8.2 Python community module

Part of Symbolica community
Import Idenso from `symbolica.community.idenso`. The `idenso` Cargo package supplies this community module when built with its Python feature; it is not a standalone PyPI distribution and is not included in the GammaLoop wheel merely because the crates share a repository.

Install the published assembly with `python -m pip install --upgrade symbolica`, then verify it with `python -c "import symbolica.community.idenso"`. There is no `pip install idenso` fallback. For a source build, add this crate to the external symbolica-community assembly with the `python` feature and call `IdensoModule`'s `SymbolicaCommunityModule` registration when that extension is assembled. Building the Rust crate alone does not add the community module to an already installed Symbolica package.

The generated Python API records exact signatures and defaults. Its operations group naturally into four phases:

- setup: importing the community module registers its symbols;
- expansion: `expand_bis`, `expand_mink`, `expand_mink_bis`, `expand_metrics`, and `expand_color`;
- index preparation: `wrap_indices`, `wrap_dummies`, `list_dangling`, `cook_indices`, and `cook_function`;
- algebra: `dirac_adjoint`, `simplify_gamma`, `to_dots`, `simplify_metrics`, and `simplify_color`.

```python
from symbolica.community.idenso import list_dangling, simplify_metrics
from symbolica.community.spenso import Representation, TensorExpression, TensorName

minkowski = Representation.mink(4)
mu = minkowski("mu")
nu = minkowski("nu")
metric = TensorExpression.g(minkowski)
momentum = TensorName("p")
expression = metric(mu, nu) * momentum(mu)

external_indices = list_dangling(expression)
reduced = simplify_metrics(expression)
assert len(external_indices) == 1
assert len(list_dangling(reduced)) == 1
```

Idenso does not define a second parser syntax: the example constructs a Spenso-compatible Symbolica expression and then applies one Idenso transformation. Keep transformations separate while developing a pipeline so expression growth and convention changes remain observable.

For implementation details, start with the Rust API and the Python binding.

9 Version history

> **History coverage**
> This version of Idenso is 0.3.0, but the published changelog currently ends at 0.2.5. Release notes for 0.3.0 are not yet available, so the history below is incomplete for that version.

9.1 Available history

The rendered Idenso history records the published history through 0.2.5 and links to its canonical source.

When upgrading Idenso, pay particular attention to representation initialization, dummy-index handling, canonicalization, and Dirac, metric, and color conventions. A rewrite change can alter the shape or ordering of a symbolic result without changing the mathematical intent, so callers that persist or compare printed expressions should verify their chosen canonical form.

9.2 Upgrade checklist

Re-run representative identities and verify initialization, dummy-index allocation, canonicalization, and printed-expression expectations.

9.3 Reproducibility record

For reproducible upgrades, record the Idenso version and enabled features with the calculation. Do not assume that changes between 0.2.5 and 0.3.0 are fully represented in the available notes.

10 Idenso versions

This page renders the canonical Idenso changelog source as part of the Idenso website, navigation, and search.

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

11.1 Unreleased

11.2 0.2.5 - 2026-01-08

11.2.1 Fixed

- fix metric initialization and update to symbolica 1.2
- fix initialize to also load ETS

11.2.2 Other

- update linnet

11.3 0.2.4 - 2025-12-15

11.3.1 Fixed

- fix canonization in presence of duals
- fix gamma normalisation
- fix warnings
- fix `add_assign` with sparse tensors
- fix projp

11.3.2 Other

- Import initialize function
- Update symbolica to 1.1.0
- update linnet
- align to atom in argument of cof for Nc
- update linnet
- add readmes
- formatting and add ref for parametric
- update to 0.17 `pyo3_stub_gen`
- update linnet and make classattr to classmethods
- proper imports and feature flags for python
- move initialize to api
- Release spenso-macros 0.3
- update to symbolica 1.0
- update sympolica to 0.20
- Properly precontract scalars
- Add back pyo3-stub-gen-derive
- update to current dev symbolica and add pattern for pslash + m conjugate
- update to linnet 0.13.1
- working canonize (up to functions) and dirac adjoint
- spenso canonize buggy
- `robust_conj` but with lots of `gamma_0s`
- first try conj
- working pow execution with wrapping
- add function generic key

11.4 0.2.2 - 2025-08-18

11.4.1 Fixed

- fix cook indices
- fix clippy errors

11.4.2 Other

- update linnet to crates
- all tests pass
- Fix Multiple Heads Bug
- update linnet
- update linnet
- update linnet
- update parse problem and add spenso:: to all symbols
- make all symbols take spenso namespace
- operate directly on coefficient list
- remove expand

11.5 0.2.1 - 2025-07-15

11.5.1 Fixed

- fix gamma algebra again

11.5.2 Other

- update versions

11.6 0.2.0 - 2025-07-15

11.6.1 Fixed

- fix `wrap_indices`
- fix gamma algebra≈

11.6.2 Other

- allow arbitrary arguments for to dots

Python API reference

The supported Python packages available in this version are listed below. The HTML manual provides a separate page for every public class, function, and constant; this printable edition keeps a compact inventory. Rust APIs use canonical Rustdoc in the HTML manual.

`idenso-community`

Exports registered by idenso.

`CanonicalizationError`

Raised when dummy-index canonicalization fails.

Raised when dummy-index canonicalization fails.

```
class CanonicalizationError(ValueError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`ColorCasimirSettings`

Immutable configuration for rewriting color invariants into a Casimir basis.

Immutable configuration for rewriting color invariants into a Casimir basis.

```
class ColorCasimirSettings:
```

Requires features: `python, python-stubs`

`rewrite_fundamental_dimension`

Kind: `Getter`

```
def rewrite_fundamental_dimension(self) -> bool:
```

Whether the fundamental dimension is rewritten with the SU(N) relation `d_F = C_A`.

`substitute_fundamental_index`

Kind: `Getter`

```
def substitute_fundamental_index(self) -> bool:
```

Whether the fundamental Dynkin index is replaced by `T_F = 1/2`.

`__new__`

Kind: `AssociatedFunction`

```
def __new__(cls, *, rewrite_fundamental_dimension: bool = True,
substitute_fundamental_index: bool = False) -> ColorCasimirSettings:
```

Configure the SU(N) dimension and fundamental-index normalizations used by Casimir rewriting.

`rewrite_fundamental_dimension`

Kind: `Parameter`

```
bool
```

Default: `True`

`substitute_fundamental_index`

Kind: `Parameter`

```
bool
```

Default: `False`

Exhaustive reference · Source

`ColorSimplifySettings`

`Immutable configuration for color simplification.`

`Immutable configuration for color simplification.`

```
class ColorSimplifySettings:
```

Requires features: `python, python-stubs`

`evaluate_traces`

Kind: `Getter`

```
def evaluate_traces(self) -> bool:
```

`Whether closed color chains are evaluated as traces.`

`expand_cross_chain_fierz`

Kind: `Getter`

```
def expand_cross_chain_fierz(self) -> bool:
```

`Whether generators on different open chains are expanded with the Fierz identity.`

`substitute_cof_dimension_invariants`

Kind: `Getter`

```
def substitute_cof_dimension_invariants(self) -> bool:
```

`Whether supported `cof(N)` invariants are replaced by explicit dimension formulas.`

`__new__`

Kind: `AssociatedFunction`

```
def __new__(cls, *, evaluate_traces: bool = True, expand_cross_chain_fierz: bool =
True, substitute_cof_dimension_invariants: bool = False) -> ColorSimplifySettings:
```

`Configure color-trace evaluation, cross-chain Fierz expansion, and invariant substitution.`

`evaluate_traces`

Kind: `Parameter`

`bool`

Default: `True`

`expand_cross_chain_fierz`

Kind: `Parameter`

`bool`

Default: `True`

`substitute_cof_dimension_invariants`

Kind: `Parameter`

`bool`

Default: `False`

Exhaustive reference · Source

CookSettings

Immutable configuration for cooking symbolic functions and index payloads.

Immutable configuration for cooking symbolic functions and index payloads.

```
class CookSettings:
```

Requires features: `python, python-stubs`

mode

Kind: Getter

```
def mode(self) -> CookMode:
```

Symbol-encoding mode.

source_filter

Kind: Getter

```
def source_filter(self) -> CookSourceFilter:
```

Function occurrences selected for cooking.

output_tags

Kind: Getter

```
def output_tags(self) -> list[str]:
```

Explicit tags attached to newly created cooked symbols.

preserve_tags

Kind: Getter

```
def preserve_tags(self) -> bool:
```

Whether matching input tags are preserved on cooked symbols.

__new__

Kind: AssociatedFunction

```
def __new__(cls, *, mode: Optional[CookMode] = None, source:
Optional[CookSourceFilter] = None, output_tags: Optional[Sequence[str]] = None,
preserve_tags: bool = False) -> CookSettings:
```

Configure how functions are selected, encoded, and tagged when cooked.

Tags must be fully namespaced Symbolica tags, for example `idenso::cooked`.
Selecting `ReversibleEncoding` changes only the encoding; use `reversible()` to also select the conventional `idenso::cooked` output tag.
`mode=None` selects `CookMode.FlattenedSymbol`.

mode

Kind: Parameter

```
Optional[CookMode]
```

Default: `None`

source

Kind: Parameter

```
Optional[CookSourceFilter]
```

Default: `None`

`output_tags`
Kind: `Parameter`

`Optional[Sequence[str]]`

Default: `None`

`preserve_tags`
Kind: `Parameter`

`bool`

Default: `False`

`flattened`
Kind: `AssociatedFunction`

```
def flattened() -> CookSettings:
```

Cook all functions into readable flattened names.

`indices`
Kind: `AssociatedFunction`

```
def indices() -> CookSettings:
```

Cook only nested function payloads inside representation indices and preserve tags.

`reversible`
Kind: `AssociatedFunction`

```
def reversible() -> CookSettings:
```

Cook all functions into stable symbols that can be restored by `uncook`.

Exhaustive reference · Source

`CookSourceFilter`
Selects the function occurrences or representation-index payloads to cook.

Selects the function occurrences or representation-index payloads to cook.

```
class CookSourceFilter:
```

Requires features: `python, python-stubs`

`any_function`
Kind: `AssociatedFunction`

```
def any_function() -> CookSourceFilter:
```

Select every function-like subexpression.

`function_tags`
Kind: `AssociatedFunction`

```
def function_tags(filter: CookTagFilter) -> CookSourceFilter:
```

Select function heads accepted by `filter`.

`filter`
Kind: `Parameter`

CookTagFilter

representation_index_payload

Kind: AssociatedFunction

```
def representation_index_payload(filter: Optional[CookTagFilter] = None) ->
CookSourceFilter:
```

Select only function payloads inside representation indices.

When `filter` is supplied, the payload's function head must also match it.

filter

Kind: Parameter

Optional[CookTagFilter]

Default: None

Exhaustive reference · Source

CookTagFilter

A tag predicate used to select which function heads are cooked.

A tag predicate used to select which function heads are cooked.

```
class CookTagFilter:
```

Requires features: python, python-stubs

any

Kind: AssociatedFunction

```
def any(tags: Sequence[str]) -> CookTagFilter:
```

Match a function head when it carries at least one listed tag.

tags

Kind: Parameter

Sequence[str]

all

Kind: AssociatedFunction

```
def all(tags: Sequence[str]) -> CookTagFilter:
```

Match a function head only when it carries every listed tag.

tags

Kind: Parameter

Sequence[str]

matched_output_tags

Kind: AssociatedFunction

```
def matched_output_tags() -> CookTagFilter:
```

Match the explicit output tags configured on the associated `CookSettings`.

Exhaustive reference · Source

CookingError

Raised when a symbolic function or representation index cannot be cooked.

Raised when a symbolic function or representation index cannot be cooked.

```
class CookingError(TypeError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

DiracAdjointError

Raised when a Dirac adjoint cannot be constructed consistently.

Raised when a Dirac adjoint cannot be constructed consistently.

```
class DiracAdjointError(ValueError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

DotExpansionError

Raised when compact dot notation cannot be expanded into a tensor expression.

Raised when compact dot notation cannot be expanded into a tensor expression.

```
class DotExpansionError(ValueError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

GammaConjugationError

Raised when conjugated gamma matrices cannot be rewritten consistently.

Raised when conjugated gamma matrices cannot be rewritten consistently.

```
class GammaConjugationError(ValueError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

GammaSimplifySettings

Immutable configuration for gamma-chain simplification.

Immutable configuration for gamma-chain simplification.

```
class GammaSimplifySettings:
```

Requires features: `python, python-stubs`

chain_ordering

Kind: `Getter`

```
def chain_ordering(self) -> GammaChainOrdering:
```

Ordering strategy used for open gamma chains.

evaluate_traces

Kind: `Getter`

```
def evaluate_traces(self) -> bool:
```

Whether closed gamma chains are evaluated as traces.

expand_three_gamma_epsilon

Kind: Getter

```
def expand_three_gamma_epsilon(self) -> bool:
```

Whether three four-dimensional gammas expand into a gamma5-epsilon basis.

__new__

Kind: AssociatedFunction

```
def __new__(cls, *, chain_ordering: Optional[GammaChainOrdering] = None,
evaluate_traces: bool = True, expand_three_gamma_epsilon: bool = False) ->
GammaSimplifySettings:
```

Configure gamma-chain ordering, trace evaluation, and the optional 4D three-gamma identity.

`chain_ordering=None` selects `GammaChainOrdering.RepeatedPairs`.

chain_ordering

Kind: Parameter

```
Optional[GammaChainOrdering]
```

Default: `None`

evaluate_traces

Kind: Parameter

```
bool
```

Default: `True`

expand_three_gamma_epsilon

Kind: Parameter

```
bool
```

Default: `False`

repeated_pairs

Kind: AssociatedFunction

```
def repeated_pairs() -> GammaSimplifySettings:
```

Use FORM-like repeated-pair ordering and evaluate closed traces.

canonical

Kind: AssociatedFunction

```
def canonical() -> GammaSimplifySettings:
```

Canonically order open gamma chains and evaluate closed traces.

Exhaustive reference · Source

NetworkToolingError

Raised when a symbolic tensor network cannot be parsed or evaluated.

Raised when a symbolic tensor network cannot be parsed or evaluated.

```
class NetworkToolingError(ValueError):
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

SchoonschipSettings

Immutable configuration for expression and network Schoonschip passes.

Immutable configuration for expression and network Schoonschip passes.

```
class SchoonschipSettings:
```

Requires features: `python, python-stubs`

depth_limit

Kind: Getter

```
def depth_limit(self) -> Optional[int]:
```

Maximum parsing or recursion depth, or `None` for no limit.

mode

Kind: Getter

```
def mode(self) -> SchoonschipMode:
```

Whether the pass is single-pass or recursive.

traversal

Kind: Getter

```
def traversal(self) -> Optional[SchoonschipTraversal]:
```

Recursive traversal order, or `None` in single-pass mode.

expand_contracted_sums

Kind: Getter

```
def expand_contracted_sums(self) -> bool:
```

Whether contracted sums are expanded before network execution.

simplify_chain_like_functions

Kind: Getter

```
def simplify_chain_like_functions(self) -> bool:
```

Whether chain-like function payloads are simplified recursively.

schoonschip_rank1_tensors

Kind: Getter

```
def schoonschip_rank1_tensors(self) -> bool:
```

Whether rank-one tensors participate in the Schoonschip pass.

contraction_order

Kind: Getter

```
def contraction_order(self) -> SchoonschipContractionOrder:
```

Heuristic used to choose network contractions.

__new__

Kind: AssociatedFunction

```
def __new__(cls, *, depth_limit: Optional[int] = 1, mode: Optional[SchoonschipMode] =
None, traversal: Optional[SchoonschipTraversal] = None, expand_contracted_sums: bool
```

= False, simplify_chain_like_functions: bool = False, schoonschip_rank1_tensors: bool = True, contraction_order: Optional[SchoonschipContractionOrder] = None) -> SchoonschipSettings:

Configure traversal, depth, shorthand expansion, and network-contraction policies.

`depth_limit=None` removes the recursion-depth limit. `traversal` is ignored in `SinglePass` mode.
`mode=None`, `traversal=None`, and `contraction_order=None` select `Recursive`, `BreadthFirst`, and `SmallestDegree`, respectively.

depth_limit

Kind: Parameter

Optional[int]

Default: 1

mode

Kind: Parameter

Optional[SchoonschipMode]

Default: None

traversal

Kind: Parameter

Optional[SchoonschipTraversal]

Default: None

expand_contracted_sums

Kind: Parameter

bool

Default: False

simplify_chain_like_functions

Kind: Parameter

bool

Default: False

schoonschip_rank1_tensors

Kind: Parameter

bool

Default: True

contraction_order

Kind: Parameter

Optional[SchoonschipContractionOrder]

Default: None

partial

Kind: AssociatedFunction

```
def partial() -> SchoonschipSettings:
```

Apply the default shallow recursive expression pass without rank-one tensors.

full
Kind: AssociatedFunction

```
def full() -> SchoonschipSettings:
```

Apply one unrestricted-depth pass and include rank-one tensors.

default_network
Kind: AssociatedFunction

```
def default_network() -> SchoonschipSettings:
```

Use the settings applied by `schoonschip_net` when no settings are supplied.

depth_first
Kind: AssociatedFunction

```
def depth_first(depth_limit: Optional[int] = None) -> SchoonschipSettings:
```

Recursively simplify each branch before visiting its siblings.

depth_limit
Kind: Parameter

```
Optional[int]
```

Default: None

breadth_first
Kind: AssociatedFunction

```
def breadth_first(depth_limit: Optional[int] = None) -> SchoonschipSettings:
```

Recursively simplify all branches one level at a time.

depth_limit
Kind: Parameter

```
Optional[int]
```

Default: None

single_pass
Kind: AssociatedFunction

```
def single_pass(depth_limit: Optional[int] = None) -> SchoonschipSettings:
```

Visit each eligible expression once, subject to `depth_limit`.

depth_limit
Kind: Parameter

```
Optional[int]
```

Default: None

Exhaustive reference · Source

CookMode
Selects how cooked function payloads are represented as symbols.

Selects how cooked function payloads are represented as symbols.

Available values are `FlattenedSymbol` for readable names and `ReversibleEncoding` for a
stable encoding that can later be restored by `uncook`.

class CookMode(enum.Enum):

Requires features: python, python-stubs

`FlattenedSymbol`

Kind: Variant

Build a readable symbol name from the function name and its arguments.

`ReversibleEncoding`

Kind: Variant

Store a stable encoding that can later be restored by `uncook` with matching settings.

Exhaustive reference · Source

`GammaChainOrdering`

Controls how open gamma chains are reordered during simplification.

Controls how open gamma chains are reordered during simplification.

Available values are `RepeatedPairs`, which only moves matching matrices together, and
`Canonical`, which canonically orders the complete open chain.

class GammaChainOrdering(enum.Enum):

Requires features: python, python-stubs

`RepeatedPairs`

Kind: Variant

Move repeated gamma matrices toward each other without reordering unrelated factors.

`Canonical`

Kind: Variant

Canonically order open chains using adjacent Clifford-algebra swaps.

Exhaustive reference · Source

`SchoonschipContractionOrder`

Selects the heuristic used to choose the next tensor-network contraction.

Selects the heuristic used to choose the next tensor-network contraction.

Available values are `SmallestDegree`, `LargestDegree`, `MinLargestOperandBytes`, `MinProductTerms`, `MinProductBytes`, `SmallestDegreeMinLargestOperandBytes`, `SmallestDegreeMinProductTerms`, and `SmallestDegreeMinProductBytes`.

class SchoonschipContractionOrder(enum.Enum):

Requires features: python, python-stubs

SmallestDegree

Kind: Variant

Contract the pair with the fewest paired tensor slots.

LargestDegree

Kind: Variant

Contract the pair with the most paired tensor slots.

MinLargestOperandBytes

Kind: Variant

Minimize the larger operand's estimated memory footprint.

MinProductTerms

Kind: Variant

Minimize the estimated number of terms in the product.

MinProductBytes

Kind: Variant

Minimize the product's estimated memory footprint.

SmallestDegreeMinLargestOperandBytes

Kind: Variant

Minimize the paired-slot count first, then the larger operand's estimated bytes.

SmallestDegreeMinProductTerms

Kind: Variant

Minimize the paired-slot count first, then the estimated product term count.

SmallestDegreeMinProductBytes

Kind: Variant

Minimize the paired-slot count first, then the estimated product bytes.

Exhaustive reference · Source

SchoonschipMode

Selects whether a Schoonschip pass runs once or recursively.

Selects whether a Schoonschip pass runs once or recursively.

Available values are `SinglePass` and `Recursive`.

class SchoonschipMode(enum.Enum):

Requires features: python, python-stubs

SinglePass

Kind: Variant

Visit each eligible expression at most once.

Recursive

Kind: Variant

Repeat traversal until the configured depth or a fixed point is reached.

Exhaustive reference · Source

SchoonschipTraversal

Selects the recursive traversal order for a Schoonschip pass.

Selects the recursive traversal order for a Schoonschip pass.

Available values are `DepthFirst` and `BreadthFirst`.

```
class SchoonschipTraversal(enum.Enum):
```

Requires features: `python, python-stubs`

DepthFirst

Kind: `Variant`

Fully simplify each branch before advancing to its siblings.

BreadthFirst

Kind: `Variant`

Advance all branches one level before descending further.

Exhaustive reference · Source

alias_subtensors

Replace nested tensor subexpressions by generated aliases.

Replace nested tensor subexpressions by generated aliases.

Returns the rewritten root followed by sorted `(alias, original)` pairs.

```
def alias_subtensors(expression: Expression, tensor_name: str) -> tuple[Expression,
list[tuple[Expression, Expression]]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

canonize

Canonically order tensor factors and deterministically rename contracted indices.

Canonically order tensor factors and deterministically rename contracted indices.

Raises `CanonicalizationError` when the expression cannot be parsed or canonicalized.

```
def canonize(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

chainify

Rewrite tensors with two slots in `representation` as explicit open-chain factors.

Rewrite tensors with two slots in `representation` as explicit open-chain factors.

```
def chainify(expression: Expression, representation: Representation | Expression) ->
Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

collect_chains

Join adjacent open chains for the supplied representation.

Join adjacent open chains for the supplied representation.

```
def collect_chains(expression: Expression, representation: Representation |
Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

collect_color

Factor an expression around tensors carrying fundamental, antifundamental, or adjoint color.

Factor an expression around tensors carrying fundamental, antifundamental, or adjoint color.

```
def collect_color(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

collect_color_constants

Factor an expression around recognized scalar color invariants such as Casimirs and indices.

Factor an expression around recognized scalar color invariants such as Casimirs and indices.

```
def collect_color_constants(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

collect_gamma_chains

Convert bispinor tensors into chain/trace shorthands and join adjacent gamma chains.

Convert bispinor tensors into chain/trace shorthands and join adjacent gamma chains.

```
def collect_gamma_chains(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

conjugate_transpose

Complex-conjugate an expression and transpose tensor slots in `representation`.

Complex-conjugate an expression and transpose tensor slots in `representation`.

```
def conjugate_transpose(expression: Expression, representation: Representation |
Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

cook

Encode selected functions as symbols using reversible cooking by default.

```
Encode selected functions as symbols using reversible cooking by default.

Raises `CookingError` when a selected function payload cannot be encoded.
```

```
def cook(expression: Expression, settings: Optional[CookSettings] = None) ->
Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`cook_function`

```
Convert a single function call into a flattened variable symbol.
```

```
Convert a single function call into a flattened variable symbol.

Transforms a function expression with arguments into a single symbolic variable
whose name encodes both the function name and its arguments. This is the
atomic version of `cook_indices()`, operating on individual function calls
rather than complete expressions.

**Function Cooking Transform:**
- Simple function: `f(a, b)` → `f_a_b`
- Nested arguments: `tensor(rep(mu))` → `tensor_rep_mu`
- Multiple arguments: `gamma(alpha, beta, mu)` → `gamma_alpha_beta_mu`
- Complex names: `my_function(x, y)` → `my_function_x_y`


**Constraints:**
- Input must be a single function call (not sum, product, etc.)
- Arguments must be cookable (symbols, numbers, simple functions)
- Cannot cook expressions containing polynomials or complex structures

# Arguments
- `expression`: expression representing a single function call to cook

# Returns
Expression containing the flattened variable symbol.

# Raises
`TypeError` if input is not a cookable function or contains invalid argument types.

# Examples:
```python
import symbolica as sp
from symbolica.community.idenso import cook_function

# Simple function cooking
f = sp.S('f')
a, b = sp.S('a','b')

cooked = cook_function(f(a, b))
print(cooked)  # Outputs: f_a_b
```
```

```
def cook_function(expression: Expression, settings: Optional[CookSettings] = None) ->
Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

cook_indices

Convert complex nested index structures into flattened symbolic names.

Convert complex nested index structures into flattened symbolic names.

Transforms hierarchical index expressions within tensor function arguments
into simplified, flat symbolic representations. This "cooking" process is
essential for pattern matching, simplification, and computational efficiency
when dealing with complex tensor expressions.

Index Cooking Transformation:
- Nested structure: `mink(4, f(g(h(μ))))` → `mink(4, f_g_h_mu)`
- Function chains: `lorentz(up(mu))` → `lorentz(up_mu)`
- Complex arguments: `tensor(rep(dim,type(idx)))` → `tensor(rep(dim,type_idx))`

Scope:
- Only affects indices appearing as function arguments
- Preserves top-level function structure
Arguments
- `expression`: expression containing complex nested index structures

Returns
Expression with flattened, simplified index names.

Examples:
```python
from symbolica.community.spenso import TensorName, Slot, Representation
import symbolica as sp
from symbolica.community.idenso import wrap_indices, cook_indices

T = TensorName("T")
rep = Representation.euc(3)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")
x = sp.S("x")
tensor_with_args = T(x, mu, nu)  # T(x; mu, nu)
print(tensor_with_args)
print(
    cook_indices(wrap_indices(tensor_with_args.to_expression(), sp.S("wrap")))
)
```

```
def cook_indices(expression: Expression, settings: Optional[CookSettings] = None) ->
Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

dirac_adjoint

Construct the physics-aware Dirac adjoint of a tensor expression.

Construct the physics-aware Dirac adjoint of a tensor expression.

Idenso takes the symbolic complex conjugate, reverses compatible open bispinor
chains, and

```
inserts the registered `gamma0` factors required at dangling bispinor slots.
The input must use the representation-aware Spenso forms registered on import.
Raises `DiracAdjointError` when the tensor network does not define a consistent
adjoint.

# Examples
```python
>>> from symbolica.community.idenso import dirac_adjoint, list_dangling
>>> from symbolica.community.spenso import Representation, TensorName
>>> # Built-in representations are registered automatically on import.
>>> # Re-importing the module does not require explicit re-registration.
>>> bispinor = Representation.bis(4)
>>> spinor = TensorName("u")(bispinor("alpha")).to_expression()
>>> adjoint = dirac_adjoint(spinor)
>>> len(list_dangling(adjoint)) == 1
True
>>> "gamma0" in str(adjoint)
True
```

Arguments
- `expression`: a Spenso-compatible tensor expression.

Returns
The representation-aware Dirac adjoint.
```

```
def dirac_adjoint(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

expand_bis

```
Expand products around factors carrying registered bispinor indices.
```

```
Expand products around factors carrying registered bispinor indices.

# Arguments
- `expression`: a factorized Spenso-compatible expression.

# Returns
`(structure, coefficient)` pairs distributed around bispinor-bearing factors. No
explicit
spinor components are substituted.

# Examples
```python
>>> from symbolica.community.idenso import expand_bis
>>> from symbolica.community.spenso import Representation, TensorName
>>> # Built-in representations are registered automatically on import.
>>> bispinor = Representation.bis(4)
>>> alpha, beta = bispinor("alpha"), bispinor("beta")
>>> u, v, w = TensorName("u"), TensorName("v"), TensorName("w")
>>> u_alpha = u(alpha).to_expression()
>>> v_beta, w_beta = v(beta).to_expression(), w(beta).to_expression()
>>> factorized = u_alpha * (v_beta + w_beta)
>>> terms = expand_bis(factorized)
>>> sum(structure * coefficient for structure, coefficient in terms) == u_alpha *
```
```

```
v_beta + u_alpha * w_beta
True
```

```
def expand_bis(expression: Expression) -> list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`expand_color`

Expand products around registered color factors.

Expand products around registered color factors.

```
Fundamental, antifundamental, adjoint, color-chain, color-trace, and supported
invariant
factors form the selected sector. This only distributes the symbolic expression; use
`simplify_color()` separately to apply SU(N) identities.

# Arguments
- `expression`: a factorized Spenso-compatible expression.

# Returns
`(structure, coefficient)` pairs distributed around color-bearing factors.

# Examples
```python
>>> from symbolica.community.idenso import expand_color
>>> from symbolica.community.spenso import Representation, TensorExpression
>>> # Built-in representations are registered automatically on import.
>>> adjoint, fundamental = Representation.coad(8), Representation.cof(3)
>>> antifundamental = fundamental.dual()
>>> generator = TensorExpression.t(8, 3)
>>> t_a = generator(
...     adjoint("a"), fundamental("i"), antifundamental("j")
... ).to_expression()
>>> t_b = generator(
...     adjoint("b"), fundamental("k"), antifundamental("l")
... ).to_expression()
>>> t_c = generator(
...     adjoint("c"), fundamental("m"), antifundamental("n")
... ).to_expression()
>>> factorized = t_a * (t_b + t_c)
>>> terms = expand_color(factorized)
>>> sum(structure * coefficient for structure, coefficient in terms) == t_a * t_b +
t_a * t_c
True
```
```

```
def expand_color(expression: Expression) -> list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`expand_dots`

Expand compact dot products into explicit metric and indexed-vector contractions.

```
Expand compact dot products into explicit metric and indexed-vector contractions.

Raises `DotExpansionError` when a dot product does not define a valid tensor
contraction.
def expand_dots(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`expand_in_patterns`

```
Selectively expand around the supplied expression patterns.
Selectively expand around the supplied expression patterns.

Results retain the Rust API's `(structure, coefficient)` factorization.
def expand_in_patterns(expression: Expression, patterns: Sequence[Expression]) ->
list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`expand_metrics`

```
Expand products around registered metric tensors.
Expand products around registered metric tensors.

This is a structural expansion only. It neither contracts the metrics nor substitutes
a
dimension or signature; call `simplify_metrics()` separately for supported
contractions.

# Arguments
- `expression`: a factorized Spenso-compatible expression.

# Returns
`(structure, coefficient)` pairs distributed around metric factors.

# Examples
```python
>>> from symbolica.community.idenso import expand_metrics
>>> from symbolica.community.spenso import Representation, TensorExpression
>>> # Built-in representations are registered automatically on import.
>>> minkowski = Representation.mink(4)
>>> metric = TensorExpression.g(minkowski)
>>> g_mn = metric(minkowski("mu"), minkowski("nu")).to_expression()
>>> g_rs = metric(minkowski("rho"), minkowski("sigma")).to_expression()
>>> g_ab = metric(minkowski("alpha"), minkowski("beta")).to_expression()
>>> factorized = g_mn * (g_rs + g_ab)
>>> terms = expand_metrics(factorized)
>>> sum(structure * coefficient for structure, coefficient in terms) == g_mn * g_rs +
g_mn * g_ab
True
```
def expand_metrics(expression: Expression) -> list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

expand_mink

Expand products around factors carrying registered Minkowski indices.

Expand products around factors carrying registered Minkowski indices.

This is a selective symbolic expansion: Minkowski-bearing factors become polynomial variables while unrelated sectors remain coefficients. It does not substitute explicit
four-vector components or choose a metric signature.

Arguments
- `expression`: a factorized Spenso-compatible expression.

Returns
`(structure, coefficient)` pairs distributed around Minkowski-bearing factors.

Examples
```python
>>> from symbolica.community.idenso import expand_mink
>>> from symbolica.community.spenso import Representation, TensorName
>>> # Built-in representations are registered automatically on import.
>>> minkowski = Representation.mink(4)
>>> mu, nu = minkowski("mu"), minkowski("nu")
>>> p, q, r = TensorName("p"), TensorName("q"), TensorName("r")
>>> p_mu = p(mu).to_expression()
>>> q_nu, r_nu = q(nu).to_expression(), r(nu).to_expression()
>>> factorized = p_mu * (q_nu + r_nu)
>>> terms = expand_mink(factorized)
>>> sum(structure * coefficient for structure, coefficient in terms) == p_mu * q_nu + p_mu * r_nu
True
```

```
def expand_mink(expression: Expression) -> list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

expand_mink_bis

Expand products around factors carrying Minkowski or bispinor indices.

Expand products around factors carrying Minkowski or bispinor indices.

This combines the selection patterns of `expand_mink()` and `expand_bis()` in one coefficient pass. Other representation families remain in the coefficient sector.

Arguments
- `expression`: a factorized Spenso-compatible expression.

Returns
`(structure, coefficient)` pairs distributed around both selected representation families.

Examples
```python
>>> from symbolica.community.idenso import expand_mink_bis
```

```
>>> from symbolica.community.spenso import Representation, TensorName
>>> # Built-in representations are registered automatically on import.
>>> minkowski, bispinor = Representation.mink(4), Representation.bis(4)
>>> p_mu = TensorName("p")(minkowski("mu")).to_expression()
>>> q_mu = TensorName("q")(minkowski("mu")).to_expression()
>>> u_a = TensorName("u")(bispinor("a")).to_expression()
>>> v_a = TensorName("v")(bispinor("a")).to_expression()
>>> factorized = (p_mu + q_mu) * (u_a + v_a)
>>> expected = p_mu * u_a + p_mu * v_a + q_mu * u_a + q_mu * v_a
>>> terms = expand_mink_bis(factorized)
>>> sum(structure * coefficient for structure, coefficient in terms) == expected
True
```

```
def expand_mink_bis(expression: Expression) -> list[tuple[Expression, Expression]]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`list_dangling`

```
Lists the dangling (external, uncontracted) indices present in the expression.
```

Lists the dangling (external, uncontracted) indices present in the expression.

```
Identifies and returns all indices that are not summed over (i.e., not dummy
indices). These are the "free" indices that appear in the final result and
determine the tensor rank of the expression. For dualizable representations,
downstairs indices are represented wrapped in `dind(...)`.

This is essential for:
- Verifying index conservation in tensor equations
- Determining the rank and structure of tensor expressions
- Debugging index contractions

# Arguments
- `expression`: tensor expression to analyze

# Returns
A list of expressions, each representing a free (dangling) index.

# Raises
`ValueError` when the expression cannot be parsed as a tensor network.

# Examples:
```python
from symbolica.community.spenso import TensorName, Slot, Representation
import symbolica as sp
from symbolica.community.idenso import (
    list_dangling,
)

T = TensorName("T")
rep = Representation.euc(3)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")
x = sp.S("x")
```
```

```
tensor_with_args = T(x, mu, nu, nu)  # T(x; mu, nu, nu)
# print(tensor_with_args)
print(list_dangling(tensor_with_args.to_expression()))
```

```
def list_dangling(expression: Expression) -> list[Expression]:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`metric_shorthand_to_dot`

Replace metric shorthand such as `g(p(rep), q(rep.dual()))` by `dot(p(rep), q(rep.dual()))`.

Replace metric shorthand such as `g(p(rep), q(rep.dual()))` by `dot(p(rep), q(rep.dual()))`.

```
def metric_shorthand_to_dot(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`normalize_chains`

Convert chains whose endpoints coincide into trace shorthands.

Convert chains whose endpoints coincide into trace shorthands.

```
def normalize_chains(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`normalize_dots`

Canonicalize compact dot-product shorthands without expanding their tensor structure.

Canonicalize compact dot-product shorthands without expanding their tensor structure.

```
def normalize_dots(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`schoonschip`

Simplify tensor shorthands using the configured Schoonschip traversal.

Simplify tensor shorthands using the configured Schoonschip traversal.

```
def schoonschip(expression: Expression, settings: Optional[SchoonschipSettings] =
None) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`schoonschip_net`

Parse and contract a symbolic tensor network using Schoonschip rules.

Parse and contract a symbolic tensor network using Schoonschip rules.

Set `expand_contracted_sums=True` to distribute sums before contracted products are executed.

```
Raises `NetworkToolingError` when the expression is not a valid tensor network or
contraction
fails.
```

```
def schoonschip_net(expression: Expression, settings: Optional[SchoonschipSettings] =
None, *, expand_contracted_sums: bool = False) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

simplify_color

```
Simplify registered Spenso color chains, traces, generators, and structure constants.
```

```
Simplify registered Spenso color chains, traces, generators, and structure constants.

With the default Python settings, the simplifier evaluates supported closed traces
and
expands contractions between generators on separate fundamental chains. Its
normalization
conventions are

- `Tr(T^a T^b) = TR δ^{ab}`;
- `Σ_a (T^a)_i^j (T^a)_k^l = TR (δ_i^l δ_k^j - δ_i^j δ_k^l/Nc)`;
- `Σ_a (T^a)_i^j (T^a)_j^k = CF δ_i^k`;
- `Σ_{c,d} f^{acd} f^{bcd} = CA δ^{ab}`.
Antisymmetry and Jacobi identities apply to the registered structure constants.

`CA = Nc`, `CF = (Nc² - 1)/(2Nc)`, and `TR = 1/2` are the conventional fundamental
SU(Nc) specialization, not identities imposed on every input. The default simplifier
keeps
representation invariants symbolic where possible; explicit dimension substitution is
a
separate `ColorSimplifySettings` option.

# Examples
```python
>>> from symbolica import E
>>> from symbolica.community.idenso import simplify_color
>>> # Built-in representations are registered automatically on import.
>>> generators = E('''
...     t(coad(Nc^2-1,a),cof(Nc,i),dind(cof(Nc,j)))
...     * t(coad(Nc^2-1,a),cof(Nc,k),dind(cof(Nc,l)))
... ''', default_namespace="spenso")
>>> simplified = simplify_color(generators)
>>> "t(" not in str(simplified) and "g(" in str(simplified)
True
```

Representation invariants:
Use `Representation.dimension`, `.casimir()`, `.dynkin_index()`, and `.gram(...)`
to construct the scalar invariants associated with explicitly typed color structures.

Arguments
- `expression`: expression containing SU(N) color structures

Returns
The simplified expression, reduced to representation-owned scalar invariants when
```

possible.
Unsupported or open indexed structures may remain explicitly in the result; their presence
is not an error.

Notes
Only representation-aware Spenso color forms are recognized. Plain Symbolica functions with
similar names are left unchanged.

def simplify_color(expression: Expression, settings: Optional[ColorSimplifySettings] = None) -> Expression:

Requires features: python, python-stubs

Exhaustive reference · Source

simplify_epsilon

Simplify Levi-Civita/metric contractions and pairs of Levi-Civita tensors to a fixed point.

Simplify Levi-Civita/metric contractions and pairs of Levi-Civita tensors to a fixed point.

def simplify_epsilon(expression: Expression) -> Expression:

Requires features: python, python-stubs

Exhaustive reference · Source

simplify_gamma

Simplify registered Spenso gamma chains and traces with Idenso's default rules.

Simplify registered Spenso gamma chains and traces with Idenso's default rules.

The dimension-generic part applies compatible Clifford anticommutation, adjacent contractions, and ordinary odd/even trace recursion. Chisholm identities, gamma-five anticommutation and traces, gamma-zero conjugation, and chiral-projector rules are applied
only when the expression carries explicit four-dimensional Minkowski and bispinor representations.

This function does not select or implement a dimensional-regularization gamma-five scheme.
Its gamma-five rules are strictly four-dimensional, and the default Python entry point does
not enable the optional three-gamma epsilon expansion available through `GammaSimplifySettings`.
Gamma factors must use the Spenso representation-aware forms registered on import; unrecognized plain Symbolica functions are left unchanged.

Examples
```python
>>> from symbolica import E
>>> from symbolica.community.idenso import simplify_gamma
>>> # Built-in representations are registered automatically on import.
>>> trace = E('''
...     gamma(bis(4,a),bis(4,b),mink(4,mu))
...     * gamma(bis(4,b),bis(4,a),mink(4,nu))
... ''', default_namespace="spenso")
```

```
>>> simplified = simplify_gamma(trace)
>>> "gamma(" not in str(simplified) and "g(" in str(simplified)
True
```

The native gamma argument order is `bis(dim,alpha), bis(dim,beta), mink(dim,mu)`:
`alpha` and `beta` are spinor indices, followed by the Lorentz index `mu`.
These forms can also be constructed through the HEP tensor library.

Arguments
- `expression`: expression containing gamma matrix products and traces

Returns
The simplified expression with gamma algebra applied.

Examples:
```python
from symbolica.community.spenso import TensorLibrary, TensorName
from symbolica.community.idenso import simplify_gamma
from symbolica import S, Expression
# Get HEP library with standard tensors
hep_lib = TensorLibrary.hep_lib()
# Access standard tensors like gamma matrices
gamma_structure = hep_lib[S("spenso::gamma")]
print(gamma_structure)
print(simplify_gamma(gamma_structure(3, 4, 7) * gamma_structure(7, 4, 3)))
```

```
def simplify_gamma(expression: Expression, settings: Optional[GammaSimplifySettings]
= None) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`simplify_gamma0`

Simplify products and linear combinations involving the time-like gamma matrix
`gamma0`.

Simplify products and linear combinations involving the time-like gamma matrix
`gamma0`.

```
def simplify_gamma0(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`simplify_gamma_conjugate`

Rewrite conjugated gamma matrices as gamma0-sandwiched matrices with fresh spinor
indices.

Rewrite conjugated gamma matrices as gamma0-sandwiched matrices with fresh spinor
indices.

Raises `GammaConjugationError` when the expression cannot be rewritten consistently.

```
def simplify_gamma_conjugate(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

simplify_metrics

Simplifies contractions involving metric tensors and identity tensors.

Simplifies contractions involving metric tensors and identity tensors.

Applies fundamental tensor algebra rules for metric and identity tensors:

Metric tensor rules:
- `gᵘᵛ pᵥ → pᵘ` (index raising/lowering)
- `gᵘᵛ gᵥρ → gᵘρ` or `δᵘρ` (metric composition)
- `gᵘᵤ → D` (dimension of spacetime)
- `ηᵘᵛ pᵥ → pᵘ` (flat metric contractions)

Identity tensor rules:
- `δᵘᵛ pᵥ → pᵘ` (Kronecker delta contraction)
- `δᵘᵤ → D` (trace of identity)

The function recognizes metrics as `spenso::g(...)`

Arguments
- `expression`: expression containing metric/identity tensor contractions

Returns
The simplified expression with metric rules applied.

Examples:
```python
from symbolica.community.idenso import simplify_metrics, to_dots
from symbolica.community.spenso import Representation, TensorExpression, TensorName
q = TensorName("q")
rep = Representation.euc(3)
g = TensorExpression.g(rep)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")
print(simplify_metrics(g("mu", "nu") * q(mu)))
```

```
def simplify_metrics(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

spenso_conjugate

Complex-conjugate an expression while keeping unevaluated conjugations explicit.

Complex-conjugate an expression while keeping unevaluated conjugations explicit.

```
def spenso_conjugate(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

to_cof_dimension_invariants

Replace supported `cof(N)` Casimir, Dynkin-index, and Gram invariants by dimension formulas.

Replace supported `cof(N)` Casimir, Dynkin-index, and Gram invariants by dimension formulas.

```
def to_cof_dimension_invariants(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

to_color_casimir

Rewrite the supplied color-representation dimensions and invariants into a Casimir basis.

Rewrite the supplied color-representation dimensions and invariants into a Casimir basis.

`fundamental` and `adjoint` accept Spynso `Representation` objects or their symbolic expressions. Only scalar coefficient positions are rewritten.

```
def to_color_casimir(expression: Expression, *, fundamental: Representation |
Expression, adjoint: Representation | Expression, settings:
Optional[ColorCasimirSettings] = None) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

to_dots

Converts contracted Lorentz/Minkowski indices into dot product notation.

Converts contracted Lorentz/Minkowski indices into dot product notation.

Automatically identifies and converts patterns like `p(mink(D, mu)) * q(mink(D, mu))` into the compact, representation-carrying `dot(p(mink(D)), q(mink(D)))` notation. This
simplification is essential for physics calculations involving four-vectors.

The function recognizes:
- Contracted vector indices: `pᵘqᵤ → p·q`
- Multiple contractions: `pᵘqᵤrᵛsᵥ → (p·q)(r·s)`
- Self-contractions: `pᵘpᵤ → p²`

Arguments
- `expression`: expression containing contracted Minkowski vector indices

Returns
The expression with vector contractions converted to dot products.

Examples:
```python
from symbolica.community.idenso import to_dots
from symbolica.community.spenso import Representation, TensorName
p = TensorName("p")
q = TensorName("q")
rep = Representation.euc(3)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")

print(to_dots( p(mu)*q(mu)))
```

```
def to_dots(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`uncook`

Restore symbols produced by matching reversible cooking settings.

Restore symbols produced by matching reversible cooking settings.

```
def uncook(expression: Expression, settings: Optional[CookSettings] = None) ->
Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`undo_all`

Expand Schoonschip, dot, chain, and trace shorthands into explicit tensor expressions.

Expand Schoonschip, dot, chain, and trace shorthands into explicit tensor expressions.

Raises `NetworkToolingError` when the expression is not a valid tensor network or evaluation fails.

```
def undo_all(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`undo_chain`

Expand open-chain shorthands while leaving dots, traces, and Schoonschip forms compact.

Expand open-chain shorthands while leaving dots, traces, and Schoonschip forms compact.

Raises `NetworkToolingError` when the expression is not a valid tensor network or evaluation fails.

```
def undo_chain(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`undo_dots`

Expand dot-product shorthands while leaving other tensor shorthands compact.

Expand dot-product shorthands while leaving other tensor shorthands compact.

Raises `NetworkToolingError` when the expression is not a valid tensor network or evaluation fails.

```
def undo_dots(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

undo_schoonschip

Expand Schoonschip tensor shorthands while leaving dots, chains, and traces compact.

Expand Schoonschip tensor shorthands while leaving dots, chains, and traces compact.

Raises `NetworkToolingError` when the expression is not a valid tensor network or evaluation fails.

```
def undo_schoonschip(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

undo_single_length

Replace one-factor chain shorthands by their underlying tensor factor.

Replace one-factor chain shorthands by their underlying tensor factor.

```
def undo_single_length(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

undo_trace

Expand trace shorthands while leaving dots, chains, and Schoonschip forms compact.

Expand trace shorthands while leaving dots, chains, and Schoonschip forms compact.

Raises `NetworkToolingError` when the expression is not a valid tensor network or evaluation fails.

```
def undo_trace(expression: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

wrap_color

Expand around color structures and wrap each resulting scalar coefficient with `symbol`.

Expand around color structures and wrap each resulting scalar coefficient with `symbol`.

```
def wrap_color(expression: Expression, symbol: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

wrap_dummies

Wraps only the dummy (contracted) indices within the expression using a header symbol.

Wraps only the dummy (contracted) indices within the expression using a header symbol.

Similar to `wrap_indices`, but selectively identifies and wraps only contracted indices (those appearing once upstairs and once downstairs, or twice in a self-dual representation), leaving external (dangling) indices untouched.

```
This is crucial for proper index management in tensor calculations.

Contracted indices are those that:
- Appear in both upper and lower positions (for dualizable reps)
- Appear twice in the same position (for self-dual reps)
- Are summed over (Einstein summation convention)

# Arguments
- `expression`: input expression containing both dummy and free indices
- `header`: symbol to use as wrapper function name for dummy indices only

# Returns
A new expression with only contracted indices wrapped.

# Raises
`ValueError` when the expression cannot be parsed as a tensor network.

# Examples:
```python
from symbolica.community.spenso import TensorName, Slot, Representation
import symbolica as sp
from symbolica.community.idenso import simplify_metrics, wrap_dummies

T = TensorName("T")
rep = Representation.euc(3)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")
x = sp.S("x")
tensor_with_args = T(x, mu, nu, nu)  # T(x; mu, nu, nu)
# print(tensor_with_args)
print(wrap_dummies(tensor_with_args.to_expression(), sp.S("wrap")))

```

def wrap_dummies(expression: Expression, header: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

`wrap_indices`

```
Wrap all abstract indices with a header symbol # Arguments - `expression`: input
expression containing tensor indices - `header`: symbol to use as the wrapper
function for all indices # Returns Expression with all indices wrapped by the header
symbol.

Wrap all abstract indices with a header symbol

# Arguments
- `expression`: input expression containing tensor indices
- `header`: symbol to use as the wrapper function for all indices

# Returns
Expression with all indices wrapped by the header symbol.

# Examples:
```python
```
```

```
from symbolica.community.spenso import TensorName, Slot, Representation
import symbolica as sp
from symbolica.community.idenso import wrap_indices

T = TensorName("T")
rep = Representation.euc(3)
# With slots (creates TensorExpression)
mu = rep("mu")
nu = rep("nu")
x = sp.S("x")
tensor_with_args = T(x, mu, nu)  # T(x; mu, nu)
print(tensor_with_args)
print(wrap_indices(tensor_with_args.to_expression(), sp.S("wrap")))

```

```
def wrap_indices(expression: Expression, header: Expression) -> Expression:
```

Requires features: `python, python-stubs`

Exhaustive reference · Source

Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** `latest`
- **Documentation route:** `/products/idenso/latest/`
- **Source revision:** `d645c0fcabf2e614eafffa0f80e6d0c9128308af`

Package versions and enabled features

- `gammaloop/gammalooprs` — `0.3.4` (`no optional features`)
- `gammaloop/gammaloop-api` — `0.1.0` (`features: cli`)
- `gammaloop/gammaloop` — `0.3.4` (`features: python, abi3, python-stubs`)
- `linnet/linnet` — `0.17.0` (`features: native, serde, bincode, drawing, symbolica`)
- `linnet/linnet-py` — `0.1.0` (`features: extension-module, abi3`)
- `spenso/spenso` — `0.6.0` (`features: native, symbolica`)
- `spenso/spenso-macros` — `0.3.2` (`no optional features`)
- `spenso/spenso-hep-lib` — `0.1.7` (`features: native`)
- `spenso/spynso3` — `0.1.2` (`features: python-stubs`)
- `idenso/idenso` — `0.3.0` (`features: native, reference-cases`)
- `idenso/idenso` — `0.3.0` (`features: python, python-stubs`)
- `vakint/vakint` — `0.1.2` (`no optional features`)
- `vakint/vakint` — `0.1.2` (`features: python, python-stubs`)